

NIST Parity Test Report — Cryogenic Material Dashboard v0.4.9

Date: 2026-05-18

Test File: tests/nist_parity.test.js

Status:  ALL 766 TESTS PASSED

Executive Summary

Comprehensive regression testing validates that all material property calculations in `js/materials.js` produce results identical to independently-implemented NIST equations. All 10 materials across all available properties (k, cp, tc) pass with zero failures.

Test Coverage

Materials Tested (10/10)

#	Material Key	Material Name	Properties	Equation Types
1	AISI316	AISI 316 Stainless Steel	k, cp, tc	polylog, piecewise-logpoly, thermal-contraction
2	Al6061T6	Aluminum 6061-T6	k, cp, tc	polylog, thermal-contraction
3	G10Normal	G-10 CR Fiber-glass Epoxy (Normal)	k, cp, tc	polylog, thermal-contraction
4	G10Warp	G-10 CR Fiber-glass Epoxy (Warp)	k, cp, tc	polylog, thermal-contraction
5	CuRRR50	OFHC Copper RRR 50	k, cp	rational, polylog
6	CuRRR100	OFHC Copper RRR 100	k, cp	rational, polylog
7	CuRRR150	OFHC Copper RRR 150	k, cp	rational, polylog
8	CuRRR300	OFHC Copper RRR 300	k, cp	rational, polylog
9	CuRRR500	OFHC Copper RRR 500	k, cp	rational, polylog
10	Ti64	Ti-6Al-4V	k, tc	polylog, thermal-contraction

Properties Tested

- **Thermal Conductivity (k):** 10 materials, 4–300 K range
- **Specific Heat (cp):** 9 materials (Ti64 excluded — no cp data), 4–300 K range
- **Thermal Contraction (tc):** 5 materials (AISI316, Al6061T6, G10Normal, G10Warp, Ti64), 4–300 K range

Test Sections & Results

Section 1: NIST Coefficient Verification (8 tests)

Transcribed NIST-published coefficients compared byte-for-byte against `data/materials.json`.

Material	Property	Status
AISI316	k coefficients	✓ Exact match
AISI316	cp piece 1 (4–50 K)	✓ Exact match
AISI316	cp piece 2 (50–300 K)	✓ Exact match
Al6061T6	k coefficients	✓ Exact match
Al6061T6	cp coefficients	✓ Exact match
G10Normal	k coefficients	✓ Exact match
G10Normal	cp coefficients	✓ Exact match
G10Warp	k coefficients	✓ Exact match

Section 2: Evaluator Parity (312 tests)

Independent NIST equation implementations compared against `propertyValue()` at 9–15 temperature points per material/property pair. Tolerance: 1×10^{-12} .

All 24 material-property combinations pass with zero relative error.

Section 3: Copper RRR Rational Model (19 tests)

Deep validation of the `sqrt(T)`-based rational evaluator for OFHC Copper.

Material	Peak T (K)	Peak k (W/m·K)	k(300K) (W/m·K)	Status
CuRRR50	26.5	1,475.8	~396	✓
CuRRR100	22.0	2,442.1	~396	✓
CuRRR150	19.5	3,247.3	~396	✓
CuRRR300	16.5	5,280.3	~396	✓
CuRRR500	14.0	7,613.6	~396	✓

Cross-RRR ordering verified: Higher RRR → higher low-T peak, as expected physically.

Section 4: Thermal Contraction Validation (13 tests)

- All 5 materials show negative contraction at low temperatures

- All materials near zero at 293 K reference temperature
- Low-T branch correctly activates for AISI316, Al6061T6, Ti64

Section 5: Edge Cases (51 tests)

- All boundary temperatures (range min/max) produce finite values
- Piecewise boundary at T=50 K for AISI316 cp confirmed
- Null correctly returned for missing properties (Ti64 cp)
- Null correctly returned for non-existent property names

Section 6: SSOT evalRational() Parity (10 tests)

Golden-value regression fixtures for CuRRR100 and CuRRR300 at 5 temperatures each. Tolerance: 1×10^{-14} .

Section 7: Physical Reasonableness (4 tests)

- Cu k(300K) = 396.3 >> SS316 k(300K) = 15.3 ✓
- G10 k(100K) = 0.31 << Al6061 k(100K) = 97.7 (insulator vs metal) ✓
- AISI316 cp(300K) = 490.2 J/(kg·K) (reasonable for steel) ✓
- cp increases with temperature (CuRRR100: 7.5 → 389.4) ✓

Section 8: Continuity Check (19 tests)

Dense sampling (200 points per curve) confirms no discontinuities (>100× step jumps) in any k or cp curve.

Equation Forms Validated

Polylog (NIST standard form)

$$\log_{10}(y) = a + b \cdot \log_{10}(T) + c \cdot (\log_{10}T)^2 + \dots + i \cdot (\log_{10}T)^8$$

$y = 10^{\text{(above)}}$

Rational (OFHC Copper k only)

$$\log_{10}(k) = (a + c \cdot \sqrt{T} + e \cdot T + g \cdot T^{(3/2)} + i \cdot T^2) / (1 + b \cdot \sqrt{T} + d \cdot T + f \cdot T^{(3/2)} + h \cdot T^2)$$

$k = 10^{\text{(above)}}$

Thermal Contraction

$$y(T) = a + bT + cT^2 + dT^3 + eT^4$$

Optional: $y = f$ **for** $T < T_{\text{low}}$

Acceptable Tolerances

Test Category	Tolerance	Justification
Evaluator parity	1×10^{-12} relative	Same coefficients, same IEEE 754 arithmetic
Golden-value regression	1×10^{-14} relative	Identical computation paths
Physical reasonableness	Domain-specific bounds	Engineering judgment per NIST ranges

NIST References

1. **NIST Cryogenic Materials Properties Database** — trc.nist.gov/cryogenics
2. **NIST Monograph 177** — “Properties of Copper and Copper Alloys at Cryogenic Temperatures” (Simon, Drexler, Reed, 1992)
3. **Marquardt, Le, Radebaugh (2000)** — “Cryogenic Material Properties Database,” 11th International Cryocooler Conference

Run Command

```
cd cryo_dashboard_v0_3_0/cryo_dashboard_v0_3_0
node tests/nist_parity.test.js
```